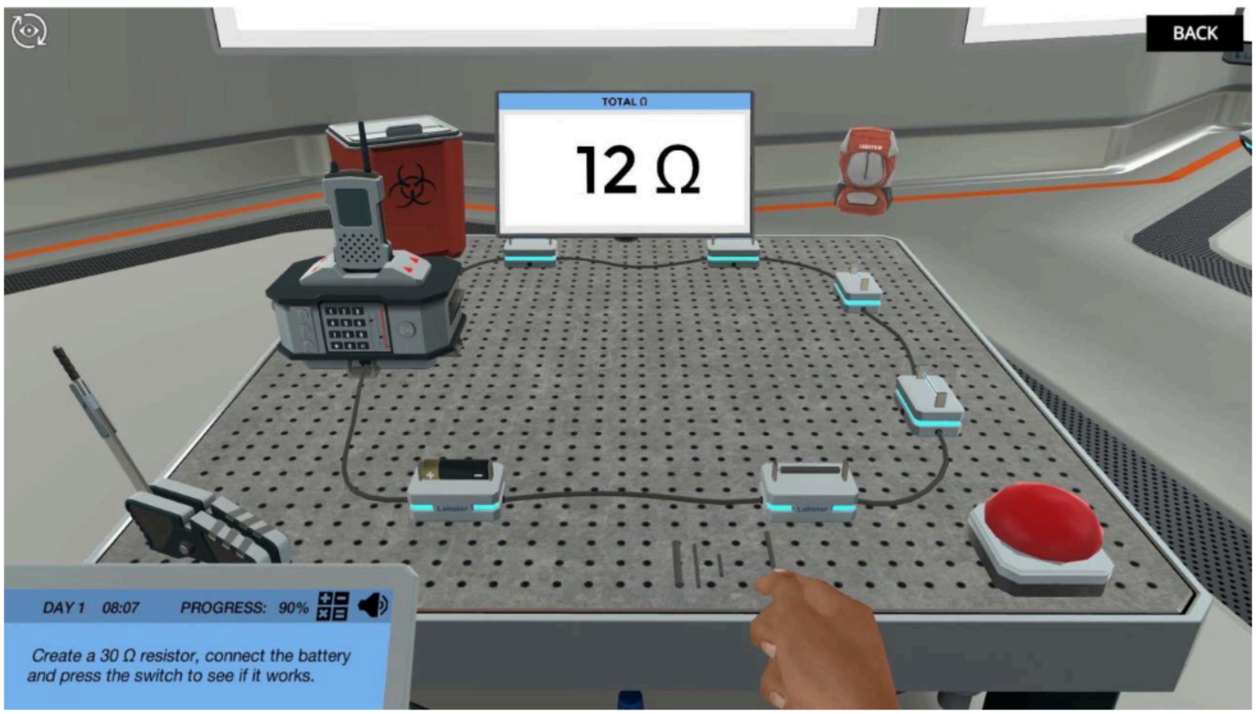
# Electrical Resistance: Apply Ohm's Law to simple circuits

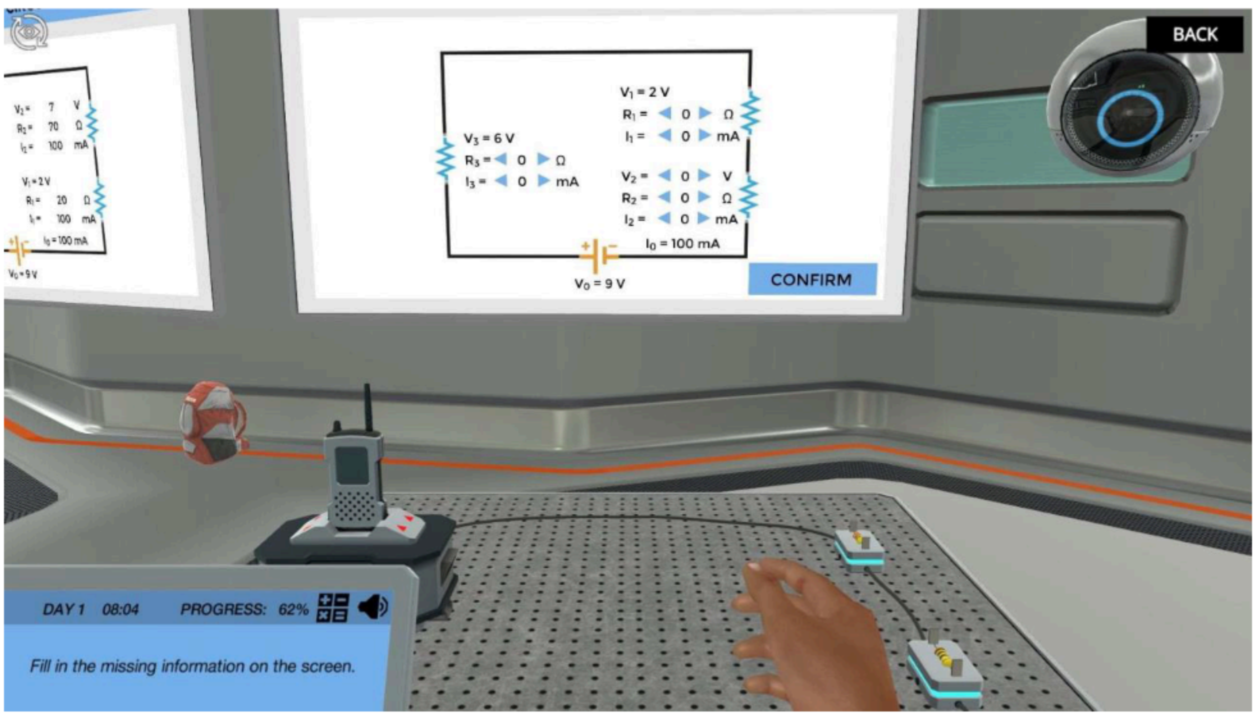
In this assignment, you will be playing an immersive online learning simulation.

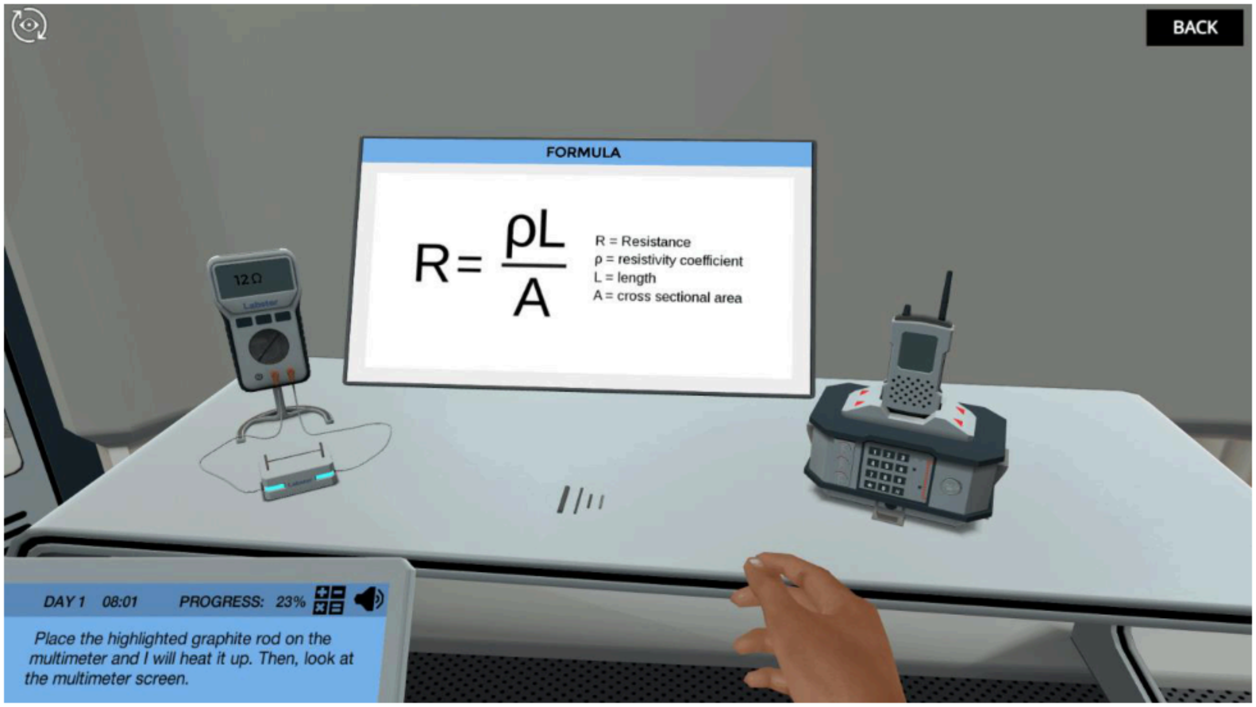
Below you can read the main learning outcomes and techniques you'll practice, as well as a brief introduction and storyline to the simulation.

Once you are ready, you can start the simulation by pressing the simulation launch button/link.




DAY 1 08:00 PROGRESS: 7%
Interact with the LabPad if you want to read the text.


BACK
V₁ = 2 V
R₁ = 0 Ω
I₁ = 0 mA
V₃ = 6 V
R₃ = 0 Ω
I₃ = 0 mA
V₂ = 0 V
R₂ = 0 Ω
I₂ = 0 mA
I₀ = 100 mA
V₀ = 9 V
CONFIRM
DAY 1 08:04 PROGRESS: 62%
Fill in the missing information on the screen.




**Learning Objectives**

*At the end of this simulation, you will be able to…*

- Define the concepts of resistivity and resistance
- Explain how resistance is affected by length, width, type of material, and temperature
- Apply Ohm's law to simple circuits
- Determine the effect of combining resistances in series and parallel
- Apply the principles of charge and energy conservation to more advanced circuits

**Techniques:**

- Circuit building

**About this Learning Simulation**

Join the resistance! In this simulation, you will learn how resistors affect electric current and how modifying the size of a resistor can modify its resistance. You will also get the chance to build your own circuit to repair a broken radio and contact your scientist colleagues.

## Repair your radio to contact scientists in Antarctica

Your radio transmitter broke down and you will need to repair it in order to contact your fellow colleagues in Antarctica. To do this, you will need to learn how resistors work and how their resistance is modified by different parameters or by how they are wired.

## Build your own circuit

You will get the chance to carry out your own experiments to see how the resistance changes when you change the length or width of a resistor. From what you've learned, you will have to replace the broken circuit in the radio and decide on how to create your own! Practice wiring your circuits with Labster's new circuit tables, where you can visualize the many different resistors you can use.

## Combine resistors and make the radio work

You will be able to test many different circuits and see how they work. Additionally, you will see how voltage, resistance and current are linked by using Ohm's Law. With this knowledge and the battery you have available, you will have to decide which resistors you would need to use. Will you be able to get back into contact with your fellow scientists in Antarctica?